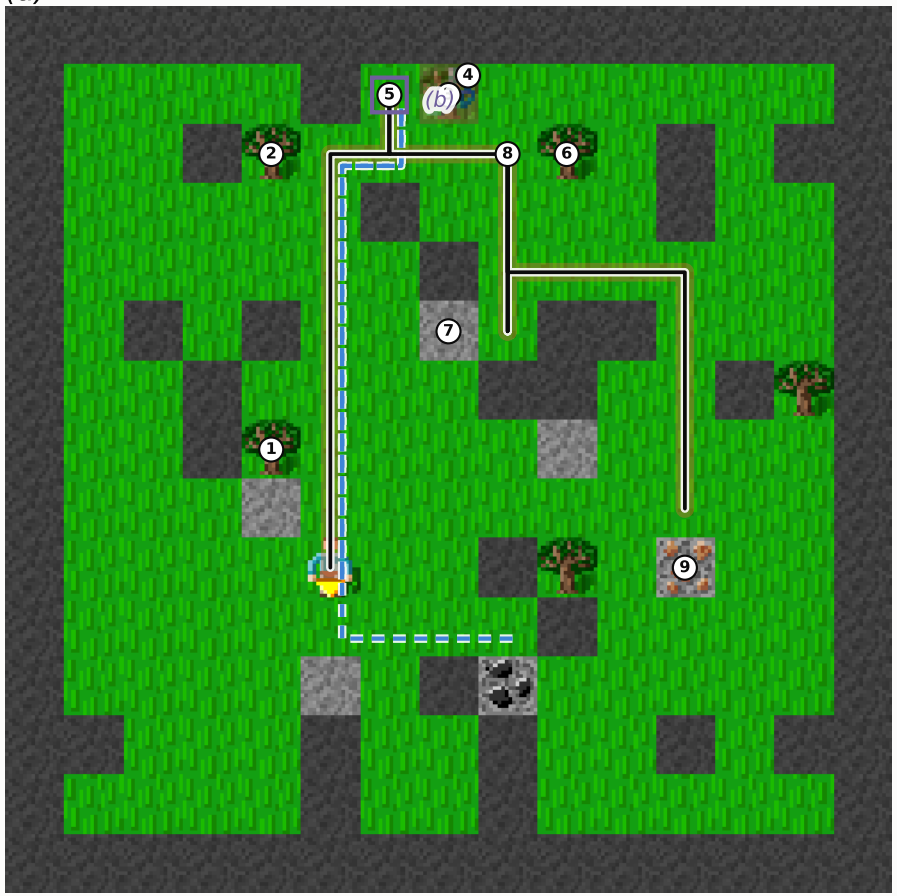
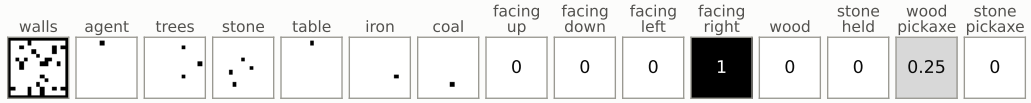
(a) held-out field 14: iron worth 1.1, coal worth 0.5, 0.05 per action
iron chain 40 actions, coal chain 32: a detour of 8 < 12, so iron



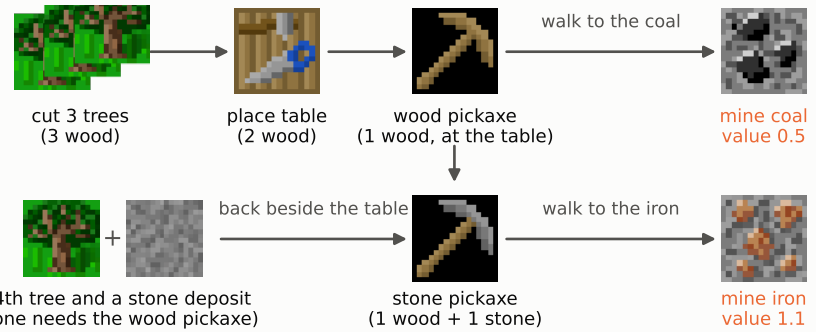
1 tree cut, step 4
2 tree cut, step 11
3 tree cut, step 15
4 table placed, step 16
5 wood pickaxe made, step 17
6 tree cut, step 21
7 stone mined, step 26
8 stone pickaxe made, step 30
9 iron mined, step 40

expert's iron plan, 40 actions — expert's coal plan, 32 actions — policy's rollout, 40 actions

(b) the observation at step 17, table down and wood pickaxe in hand: seven map planes, then facing and inventory as planes constant over the grid (counts / 4)



(c) the tech tree the expert plans over: the iron chain is the coal chain plus a detour for its ingredients



(d) the rule: iron iff its detour costs under (1.1 − 0.5) / 0.05 = 12 actions; iron on 30% of fields

